

Systematic Fine-Tuning of Transformer Models for Domain-Specific Misinformation Detection on Social Media

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Abstract

Social media platforms have become primary vectors for the dissemination of misinformation, yet the development of effective detection systems remains largely confined to English-language resources. This paper presents a systematic and transferable framework for fine-tuning pretrained transformer models to detect domain-specific deceptive content on social media, addressing the broader methodological gap in how these models are adapted for specialized tasks. The proposed framework is structured as a three-stage pipeline: (1) corpus unification—aggregating and standardizing fragmented datasets into a single balanced resource; (2) systematic regularization—a disciplined hyperparameter optimization strategy combining aggressive dropout, L2 regularization, learning rate scheduling, and early stopping to control overfitting during fine-tuning; and (3) production deployment—containerizing the optimized model as a web application for real-time inference. The framework is validated through a case study on Spanish-language fake news detection, a critically underserved domain. A unified corpus of 61,674 Spanish news articles was constructed from four heterogeneous sources, achieving near-perfect class balance (49.8% fake, 50.2% real). Through over 500 GPU hours of systematic experimentation, the proposed regularization strategy for DistilBERT (Distilled Bidirectional Encoder Representations from Transformers) achieved 95.36% accuracy with a validation-training loss gap of only 0.058, representing a 23.33 percentage point improvement over metaheuristic-optimized classical baselines. The resulting model was deployed as a Docker-containerized web application capable of real-time URL analysis. Crucially, the framework is designed to be domain-agnostic: the same pipeline can be applied to other forms of social media deception—including investment scams, fraudulent e-commerce pages, and phishing content—by substituting the training corpus. The complete framework—corpus, model, and application—is publicly available to support reproducibility and further research.

Keywords: misinformation detection; transformer fine-tuning; social media analysis; domain-specific NLP; systematic regularization; DistilBERT; transferable framework; Spanish fake news; web application deployment

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1. Introduction

The proliferation of deceptive content on social media platforms represents one of the most pressing challenges at the intersection of machine learning and digital society [1]. The rapid expansion of social networks has transformed how information is created,

disseminated, and consumed, creating unprecedented opportunities for the spread of misinformation, fraudulent schemes, and manipulative content. This phenomenon disproportionately affects vulnerable demographics with limited media literacy skills [2,3], undermining public trust in journalism, distorting democratic processes, and threatening public health outcomes [4,5]. Critically, the problem extends far beyond traditional fake news: social media platforms are increasingly exploited for investment scams—such as pages impersonating state-owned enterprises with fabricated profit schemes [6]—fraudulent e-commerce impersonations featuring fake testimonials and artificial urgency [7,8], and an alarming proliferation of fake storefronts impersonating major retail chains with fabricated liquidation sales [9–12]. Celebrity health misinformation—such as pages exploiting public figures’ names to promote fraudulent medical products [13]—further compounds the problem. These diverse forms of digital deception disproportionately target elderly and digitally illiterate populations, and their sheer volume on social media feeds underscores the urgent need for automated, scalable detection systems.

While transformer-based models have achieved state-of-the-art performance in English-language misinformation detection [14–16], a fundamental methodological gap persists: the absence of systematic, reproducible frameworks for adapting these general-purpose models to *domain-specific* deceptive content detection tasks. Most existing studies focus on reporting end-task accuracy without providing a transferable methodology that practitioners can apply to new domains, languages, or content types [17]. This gap is particularly acute for non-English languages and for emerging forms of social media fraud that lack established benchmark datasets.

The field of automated deceptive content detection has evolved through distinct methodological paradigms, reflecting advances in machine learning and natural language processing. Early research in the 2010s relied on classical machine learning approaches—Support Vector Machines (SVMs), Naïve Bayes, Random Forests—combined with hand-crafted linguistic features such as Term Frequency-Inverse Document Frequency (TF-IDF) and Bag-of-Words (BoW) representations [18–20]. While computationally efficient, these methods struggled to capture semantic nuances, sarcasm, and contextual meaning, limiting their effectiveness against sophisticated misinformation tactics.

The mid-2010s witnessed a paradigm shift toward deep learning architectures. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) enabled automatic feature extraction from text, eliminating the need for manual feature engineering [21,22]. However, these models were constrained by their inability to process long-range dependencies and bidirectional context effectively.

The introduction of transformer architectures—particularly BERT [23] and its variants—revolutionized the field by capturing bidirectional contextual representations through self-attention mechanisms [24]. Recent surveys [15,17] have documented the superiority of transformer-based approaches across multiple languages, with models such as FakeBERT [14] leveraging datasets like LIAR [25] and FakeNewsNet alongside techniques including emotion-aware multitask learning [26] and hybrid CNN-RNN architectures [21].

However, despite these architectural advances, three interconnected challenges remain largely unaddressed in the literature. First, there is a **lack of systematic fine-tuning methodologies**: most studies report results from a narrow hyperparameter search without documenting the optimization trajectory or providing reproducible regularization strategies that can be transferred to new tasks [27,28]. Second, there is a **scarcity of domain-specific resources for non-English languages**, particularly for Spanish—a language with over 500 million native speakers—where datasets are fragmented and incompatible [15,29]. Third, a **persistent deployment gap** separates academic models from practical tools [19]: while

papers report high accuracy, there is a striking absence of production-ready applications that communities can use for real-time verification.

This paper addresses these challenges by proposing a **systematic and transferable framework** for fine-tuning pretrained transformer models for domain-specific misinformation detection on social media. The framework is structured as a three-stage pipeline—corpus unification, systematic regularization, and production deployment—and is designed to be domain-agnostic: while validated here on Spanish-language fake news detection, the same methodology can be applied to detect other forms of social media deception, such as investment scams (e.g., fraudulent cryptocurrency or petroleum investment pages), fake e-commerce sites impersonating legitimate retailers, phishing campaigns, and other digital fraud targeting vulnerable populations.

A progressive research design was adopted: a baseline is first established with classical NLP methods (TF-IDF representations optimized via five metaheuristic algorithms), followed by a fine-tuned transformer model whose systematic regularization methodology is documented in full.

The main contributions of this work are:

- **A Transferable Fine-Tuning Framework:** A systematic three-stage pipeline (corpus unification, aggressive regularization, containerized deployment) is proposed for adapting pretrained transformer models to domain-specific deceptive content detection. The framework is explicitly designed to be transferable across languages, content domains, and social media platforms.
- **A Systematic Regularization Methodology:** Through over 500 GPU hours of experimentation across 30+ configurations, an aggressive regularization strategy for fine-tuning DistilBERT (Distilled Bidirectional Encoder Representations from Transformers) is documented, achieving 95.36% accuracy while maintaining a validation-training loss gap of 0.058. This represents a 23.33 percentage point improvement over metaheuristic-optimized classical methods and provides a reproducible recipe for practitioners facing similar optimization challenges.
- **A Unified Spanish Corpus:** A standardized corpus of 61,674 Spanish news articles was constructed from three academic datasets [27,29,30] enhanced with web-scraped satirical content, achieving near-perfect balance (49.8% fake, 50.2% real)—one of the largest resources for this task.
- **A Production-Ready Web Application:** The optimized model was deployed as a Docker-containerized web application capable of real-time URL analysis, demonstrating that the framework extends beyond academic benchmarks to deliver practical tools for digital self-defense [19].

The complete framework is publicly available to encourage reproducibility. Crucially, by substituting the training corpus, the same pipeline can detect investment scams, phishing pages, fraudulent e-commerce sites, and other forms of social media deception—an extensibility that constitutes the core technical contribution.

2. Related Work: Detection Paradigms and Methodological Gaps

The scholarly approach to deceptive content detection has evolved through several paradigms—from classical machine learning to deep learning architectures. This section examines the detection landscape, analyzes resources and optimization methodologies, and addresses the gap between research and deployment. Table 1 summarizes these paradigms.

Table 1. Comparison of Methodological Paradigms in Deceptive Content Detection on Social Media.

Paradigm	Core Principle	Key Studies	Strengths	Limitations
Media Literacy & Societal Impact	Address root causes through critical thinking education	Higdon [2], Higdon [3], Tsfati et al. [4]	Addresses cultural and cognitive factors	Does not provide automated detection
Classical Machine Learning	Statistical features from text (TF-IDF, BoW)	Shu et al. [18], Ali et al. [19], Thota et al. [31]	Computationally efficient, interpretable	Limited semantic understanding
Deep Learning (CNN/RNN)	Automatic feature extraction via neural networks	Nasir et al. [21], Zhou & Zafarani [22]	Better feature learning, reduced manual engineering	Limited handling of long-range dependencies
Transformers (BERT-based)	Bidirectional context via self-attention mechanisms	Singh et al. [15], Alghamdi et al. [17], Kaliyar et al. [14], Rout et al. [16]	State-of-the-art performance, captures semantic nuances	Computationally expensive, requires large datasets
Multimodal & Emotion-Aware	Integrates text, images, and emotional features	Choudhry et al. [26], Zhen & Li [32], Shang et al. [33]	Holistic analysis, improved accuracy	Complex architectures, limited Spanish resources

2.1. Language Resources for Deceptive Content Detection

A significant bottleneck for advancing deceptive content detection—whether fake news, scam pages, or fraudulent content—is the availability of large-scale, comprehensive datasets, particularly for non-English languages. Reis et al. [34] systematically evaluated supervised learning approaches for fake news detection, establishing that the quality and comprehensiveness of training datasets are critical determinants of classifier performance—a finding that is amplified in non-English settings where resources are scarce. In the Spanish-language domain, several research groups have created fake news datasets [27,29,30], but each used different annotation schemes and focused on different content types. These datasets cannot easily be combined for training because they use incompatible formats. This fragmentation problem is not unique to fake news; it is a recurring challenge whenever practitioners attempt to build detection systems for any new domain of deceptive content on social media. This work addresses this problem by proposing a systematic corpus unification methodology and demonstrating it by standardizing and merging three existing Spanish datasets into a single large corpus.

2.2. Systematic Fine-Tuning and Hyperparameter Optimization

BERT and similar transformer models achieve strong results in NLP tasks, but require careful hyperparameter tuning—a process that is rarely documented systematically in the literature. Most studies report a single final configuration without detailing the optimization trajectory, the failed configurations, or the rationale behind regularization choices [28]. This lack of transparency severely limits reproducibility and transferability to new domains. DistilBERT, for example, presents an attractive trade-off between performance and computational cost, yet a detailed exploration of its optimal fine-tuning configuration for domain-specific tasks has been largely overlooked. The critical role of hyperparameter calibration is not a new problem; it has been well-documented in other specialized NLP tasks for Spanish, such as in the medical domain [35,36], where deep learning techniques have been successfully applied to relation extraction and knowledge representation tasks, further demonstrating the broader applicability of systematic fine-tuning methodologies across NLP domains.

Concurrently, classical machine learning models have been pushed to their limits through the use of metaheuristic algorithms. For instance, approaches using genetic algorithms or particle swarm optimization have been explored to fine-tune classifiers. While foundational studies have established baselines using classical representations on fake news data [18], a direct and rigorous comparison between a systematically optimized Transformer and a suite of metaheuristic-optimized classical models on a large-scale corpus

has been notably absent from the literature. This paper aims to fill that gap by providing not only this direct comparison but also a detailed, reproducible methodology for Transformer regularization that can serve as a blueprint for practitioners in other detection domains.

2.3. Deployment of Detection Models for Social Media

A persistent issue in the academic NLP community is the “deployment gap”—the divide between models achieving high accuracy in research and the scarcity of practical tools available to the public. While numerous studies address fake news detection, very few result in production-ready applications—a gap that is even more pronounced for non-English languages and emerging forms of social media deception. In this work, a web application was developed that analyzes URLs in real-time, demonstrating how the proposed framework bridges the gap from research to deployment—a stage designed to be reusable for any detection model produced by the framework.

Table 2. Summary of key related works (Part 1): Corpus Creation & Transformer Application for Deceptive Content Detection.

Author(s) [Ref.]	Contribution	Key Finding / Performance	Limitation Addressed by This Work
Posadas-Durán et al. [29]	Created a pioneering Spanish corpus (971 articles) with a stylometric focus.	Stylometric features are useful for detection tasks, providing competitive results in IberLEF competitions with F1-scores around 0.85.	Small, isolated dataset. This work unifies it with others to create a large-scale resource.
Kaliyar et al. [14]	Proposed FakeBERT, a deep learning model combining BERT with CNN layers.	Demonstrated that integrating BERT embeddings with convolutional networks achieves state-of-the-art accuracy (98.9%) on English datasets.	English-focused, no transferable methodology. This work proposes a systematic framework applicable across languages.
Blanco-Fernández et al. [27]	Applied BERT/RoBERTa to a large, politically-focused dataset (57k articles).	Transformers achieve 90-98% accuracy on Spanish political fake news detection tasks with RoBERTa showing superior performance.	Domain-specific. This work uses a multi-domain corpus and performs systematic optimization .
Martínez-Gallego et al. [28]	Explored different BERT variants (including BETO) for Spanish fake news detection.	Spanish-specific models perform well for this classification task, with BETO achieving 94% accuracy on balanced datasets.	Lack of systematic optimization or a deployed application. This work provides a transferable framework and the deployed application .
Rout et al. [16]	Proposed an enhanced attention-based transformer model for reliable fake news detection.	Architectural modifications to the attention mechanism improve detection robustness and classification reliability.	English-focused. This work proposes a transferable framework validated on Spanish with systematic regularization.
García-Robledo et al. [36]	Applied deep learning for relation extraction in Spanish medical texts for knowledge representation.	Demonstrated that deep learning fine-tuning is effective for specialized Spanish NLP tasks beyond fake news detection.	Medical domain only. This work demonstrates a transferable framework applicable across multiple NLP domains.

Table 3. Summary of key related works (Part 2): Metaheuristic, Classical, and Hybrid Approaches.

Author(s) [Ref.]	Contribution	Key Finding / Performance	Limitation Addressed by This Work
Yildirim [37]	Hybrid multi-thread metaheuristic approach for fake news detection.	Novel metaheuristic combinations show promise for optimization tasks in NLP applications, achieving 89% accuracy on English datasets.	English-focused, limited systematic comparison. This work provides a transferable framework with comprehensive metaheuristic comparison.
Thota et al. [31]	Early deep learning approach using traditional neural networks for fake news detection.	Deep learning outperforms classical ML approaches for text classification tasks, showing 15-20% improvement over SVM baselines.	Pre-transformer era, English only. This work uses state-of-the-art transformers for Spanish.
Gravanis et al. [38]	Systematic benchmarking of classical ML classifiers and feature representations for fake news detection.	SVM and ensemble methods with TF-IDF/word embeddings achieve strongest results among 15 classifier-feature combinations.	English-only, no transformer models. This work applies systematic transformer optimization on Spanish data.

2.4. Recent Advances and Comprehensive Surveys (2020-2025)

The period from 2020 to 2025 has witnessed an explosion of scholarly attention to fake news detection, driven by the societal urgency of combating misinformation and the maturation of deep learning technologies. Several comprehensive surveys have synthesized the state of the field, providing critical perspectives that inform the approach presented herein.

Comprehensive Technical Reviews. Singh et al. [15] and Alghamdi et al. [17] conducted comprehensive surveys of fake news detection, highlighting transformer dominance and identifying the scarcity of non-English resources—particularly for Spanish—as a major bottleneck. Earlier surveys by Oshikawa et al. [39] and Hu et al. [40] provided foundational overviews of NLP and deep learning techniques, establishing the taxonomies upon which subsequent work has built.

Multimodal and Emotion-Aware Approaches. Choudhry et al. [26] introduced an emotion-aware multitask framework leveraging transfer learning, demonstrating that affective features significantly improve detection accuracy. Building on this direction, Zhen and Li [32] proposed C-STEER, a sentiment-aware framework modeling the lifecycle emotional evolution of fake news, while Shang et al. [33] introduced ETICD-Net, a multimodal network integrating emotion-topic injection with consistency modeling. These advances highlight the growing sophistication of detection architectures, though their reliance on multimodal features limits applicability to text-only scenarios—a gap that the present text-based approach can complement.

Hybrid Architectures. Nasir et al. [21] proposed a hybrid CNN-RNN model combining convolutional and recurrent layers, achieving competitive performance on English datasets, though such approaches remain limited for Spanish due to the lack of standardized corpora.

Attention-Enhanced Transformers. Rout et al. [16] proposed an enhanced attention-based transformer for reliable fake news detection, reinforcing that properly configured transformers consistently outperform alternative approaches—a finding motivating the systematic regularization proposed here. Sun and Yu [41] proposed a hierarchical detection model based on multi-task learning and adversarial training, sharing this work's emphasis on systematic optimization while focusing on architectural rather than regularization innovation.

Domain-Specific Deep Learning for Spanish NLP. The transferability of deep learning techniques to specialized Spanish-language tasks has been demonstrated by García-Robledo et al. [36] for medical relation extraction and Padilla Cuevas et al. [35] for medical question-answering. These works confirm that the fine-tuning challenges addressed here—hyperparameter sensitivity, overfitting, and the need for systematic optimization—are shared across multiple Spanish NLP applications.

Societal and Cultural Perspectives. Beyond technical methodologies, recent work has emphasized the cultural dimensions of misinformation. Tsifti et al. [4] argued that technological solutions must be complemented by media literacy interventions, while Higdon and Huff [2] and Higdon [3] provided critical thinking frameworks for conflict management and media literacy. Valencia-Arias et al. [42] identified key factors influencing young people's susceptibility to misinformation, highlighting the need for both technological and educational interventions. These perspectives reinforce a key motivation of the proposed framework: developing not only accurate models but accessible, deployable tools that empower communities to combat diverse forms of deceptive content.

Gaps Addressed by This Work. Despite these advances, two fundamental gaps persist. First, at the *methodological level*, no established framework exists for systematically adapting pretrained transformers to domain-specific deceptive content detection; existing studies

are task-specific and lack reproducible optimization documentation. Second, at the *resource level*, the Spanish-language domain remains critically underserved—the surveys by Singh et al. [15] and Alghamdi et al. [17], while comprehensive, do not address Spanish-specific challenges including dialectal variations and corpus fragmentation. The present work directly addresses both gaps through a transferable three-stage framework validated on Spanish fake news detection with demonstrated extensibility to other deception domains.

3. Materials and Methods

The methodology is structured as a transferable three-stage pipeline for adapting pretrained transformers to domain-specific deceptive content detection. While validated on Spanish fake news, each stage is domain-agnostic and reusable. As illustrated in Figure 1, the pipeline comprises: (1) **Corpus Unification**—aggregating and balancing heterogeneous datasets; (2) **Systematic Model Optimization**—comparing classical and deep learning paradigms through a documented regularization trajectory; and (3) **Production Deployment**—containerizing the optimal model for real-time inference.

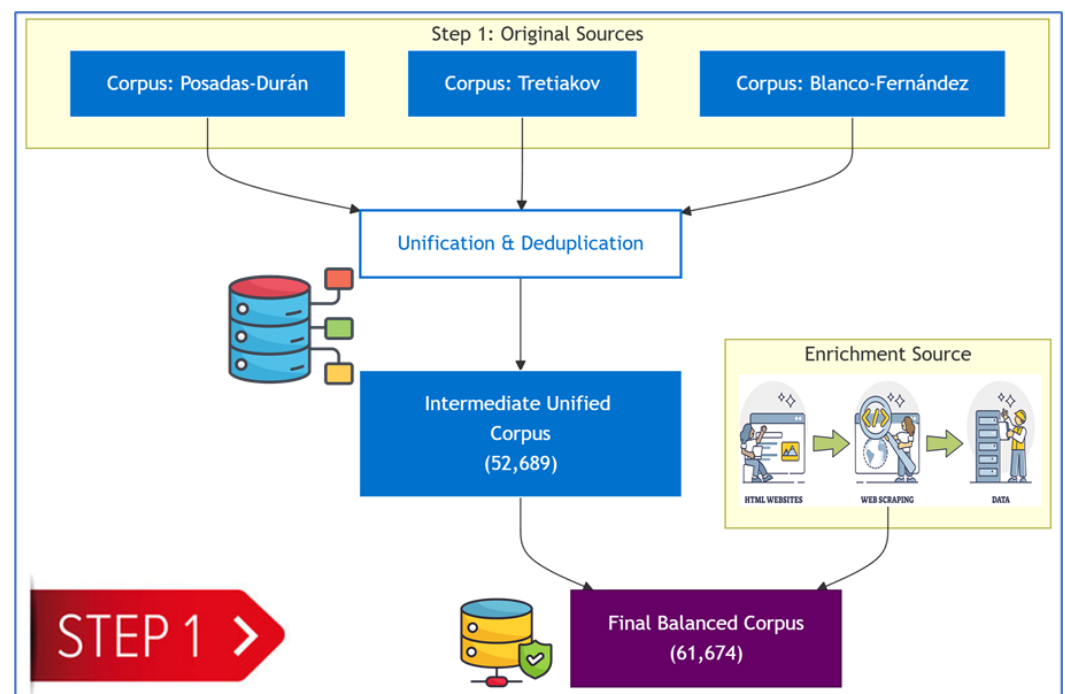


Figure 1. Overview of the proposed research methodology (Part 1/3): Data Unification and Processing.

3.1. Stage 1: Corpus Unification and Processing

The first stage addresses a common challenge: constructing a balanced dataset from fragmented sources. The unification methodology (schema standardization, content-based deduplication, targeted augmentation) is generalizable to any detection domain. A unified corpus was constructed by aggregating three publicly available datasets: the Spanish Fake News Corpus (971 articles) [29], the Tretiakov Dataset (1,958 articles) [30], and the Spanish Political Dataset (57,231 articles) [27], yielding **60,160** articles. Table 4 details each source.

Table 4. Exhaustive comparison of the characteristics of the corpora used in the construction of the unified dataset.

Comparative Aspect	Posadas-Durán	Tretiakov	Blanco-Fernández	El Deforma
Corpus Size	971 articles	1,958 articles	57,231 articles	9,000 articles
Creation Year	2019-2021	2022	2024	2025 (extraction)
Methodological Focus	Stylometric Analysis	Traditional ML	Transformer Models	Satirical Content
Thematic Domain	General	General	Political	Satirical/General
Class Distribution	Balanced	Balanced	Balanced	Fake Only
Regional Variability	Spain/LatAm	Multiple	Spain	Mexico
Annotation Quality	High	Medium	High	Automatic (inherently fake)
Contribution to Final Corpus	(multi-annotator) 1.4%	(source-based) 2.8%	(specialized) 81.2%	14.6% (added)

To ensure data quality, content hashing on the ‘text’ field identified and removed 7,712 duplicates, yielding 52,689 unique articles—a step crucial to prevent data leakage between training and testing sets. The resulting class distribution revealed a significant imbalance: 30,943 real articles (58.7%) versus 21,746 fake articles (41.3%). Rather than resorting to synthetic data generation, a targeted acquisition of satirical content was pursued from “El Deforma,” a leading Mexican satirical news site, collecting 9,000 articles. The inclusion of satire is methodologically grounded: satirical texts share key rhetorical devices with fake news—absurdity, exaggeration, and logical inconsistencies—making them an excellent proxy for training robust detection models [43]. These articles were labeled as FAKE, balancing the corpus to 61,674 total articles (49.8% fake, 50.2% real), as visualized in Figure 2.

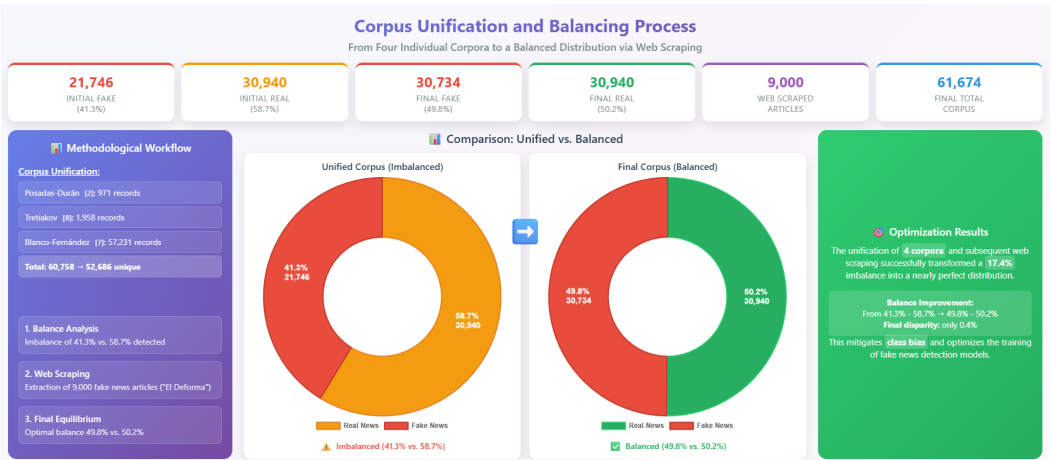


Figure 2. Visual representation of the corpus balancing process. The initial imbalanced distribution (left) was corrected by strategically adding 9,000 FAKE articles via web scraping, resulting in a nearly 50/50 final distribution (right). Source: Authors’ own elaboration based on study data.

3.2. Stage 2: Systematic Model Optimization

The second stage focuses on systematic fine-tuning documented as a reproducible optimization trajectory. Rather than reporting a single configuration, the full evolution of hyperparameter decisions is recorded (Table 6), providing a transferable recipe for practitioners. The optimization begins by establishing a classical baseline.

3.2.1. Data Partitioning and Generalization Strategy

To ensure the statistical reliability of the results and guarantee model generalization to unseen data, a rigorous data splitting protocol was implemented, as illustrated in Figure 3.

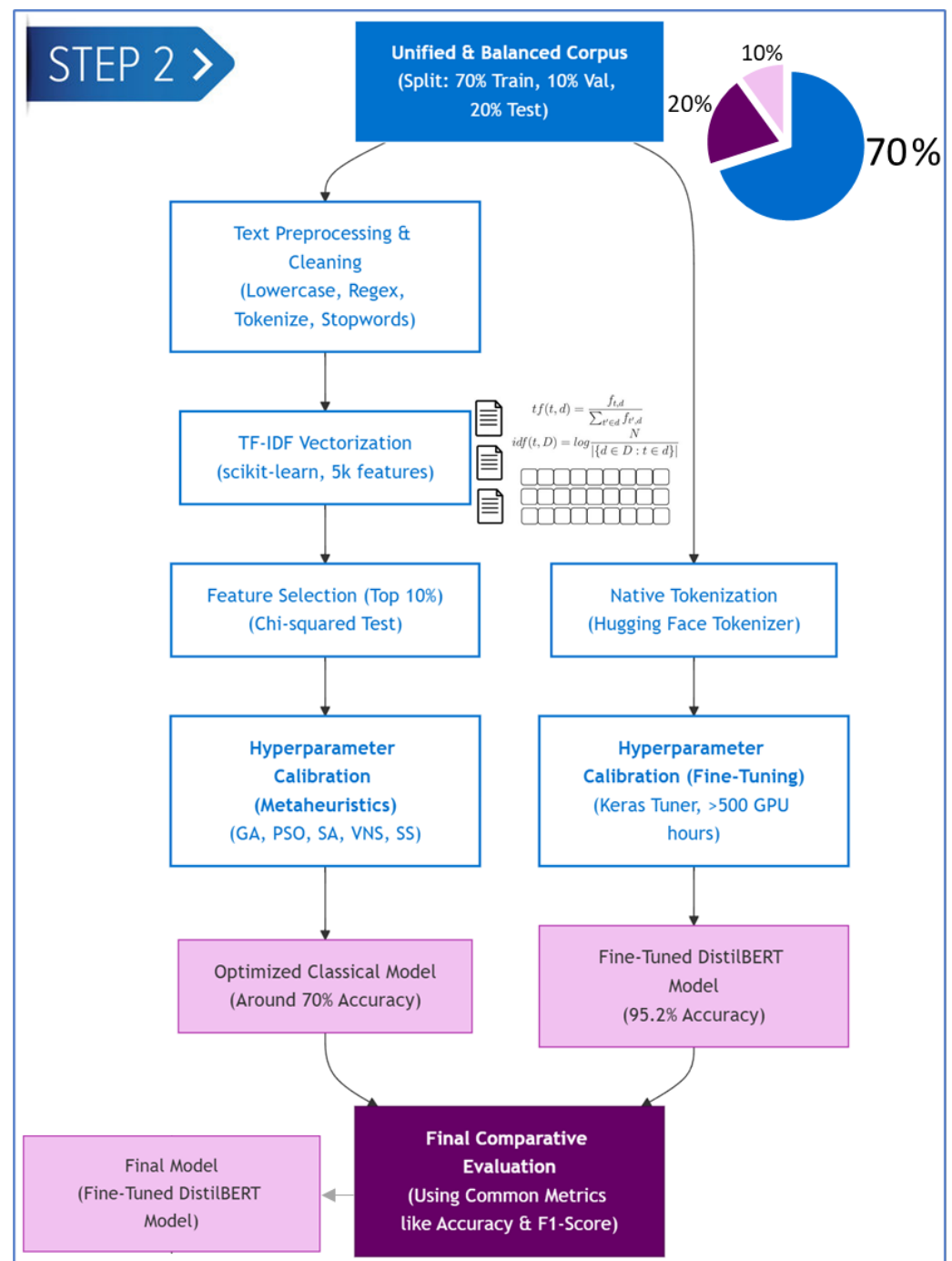


Figure 3. Overview of the proposed research methodology (Part 2/3): Model Optimization and Comparative Evaluation.

The unified balanced corpus ($N = 61,674$) was partitioned into three disjoint subsets using **stratified random sampling** to preserve the approximately 50/50 class distribution across all partitions:

- **Training Set (70%, $\approx 43,171$ articles):** Used for model fitting and gradient updates.
- **Validation Set (10%, $\approx 6,167$ articles):** Used for hyperparameter tuning, model selection, and early stopping monitoring.
- **Test Set (20%, $\approx 12,335$ articles):** Strictly isolated and used solely for the final evaluation reported in Section 4.

A “hold-out” validation strategy combined with aggressive regularization (Section 3.2.3) ensures that reported metrics reflect the model’s true generalization ability rather than memorized patterns.

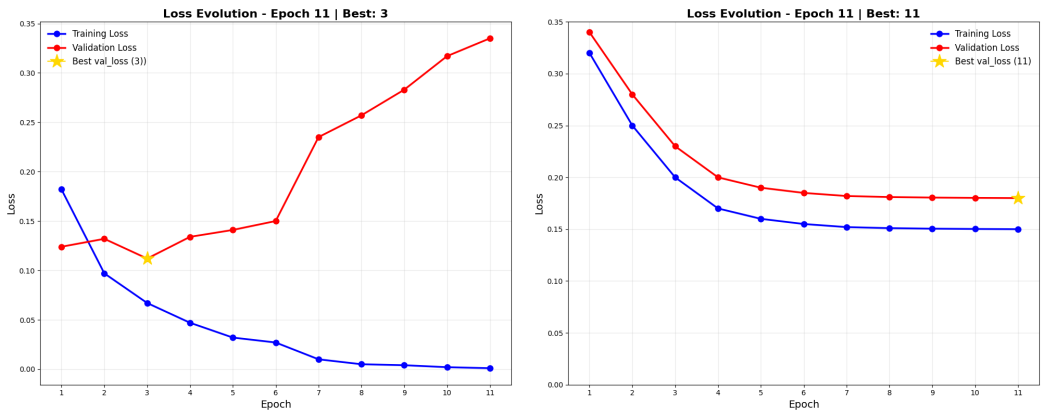
Classical Machine Learning Baseline. A baseline was established using TF-IDF feature representation [18], with dimensionality reduced to 800 features via Chi-squared test. Five metaheuristic algorithms—Multi-Start Simulated Annealing (MSA), Scatter Search (SS), Variable Neighborhood Search (VNS), Genetic Algorithm (GA), and Particle Swarm Optimization (PSO)—optimized the hyperparameters of a logistic regression model, maximizing F1-Score.

Deep Learning with Transformers. To overcome classical methods’ semantic limitations, distilbert-base-multilingual-cased was selected [44], offering the optimal trade-off between performance and efficiency—approximately four times faster than BERT with competitive accuracy (Table 5).

Table 5. Comparison of optimized BERT models for the classification task.

Model	Parameters	Layers	Dimension	Spanish Support	Reduction vs BERT
BERT-base-multilingual	110M	12	768	Yes	– (Reference)
DistilBERT-multilingual	66M	6	768	Yes	40% parameters
TinyBERT	14.5M	4	312	Limited	87% parameters

Hyperparameter Fine-tuning. Over 30 experiments and 500 GPU hours were conducted to identify configurations that mitigate overfitting—where the model memorizes training data but fails to generalize—as illustrated in Figure 4.



(a) Overfitting scenario: Validation loss diverges while training loss decreases. (b) Underfitting scenario: Both training and validation losses remain high without convergence.

Figure 4. Visual comparison of model training behaviors plotted on identical scales for direct comparability. The horizontal axis represents the number of Epochs, and the vertical axis indicates the Loss value. (a) Illustrates **overfitting**, where the model memorizes the training data (blue line decreases) but fails to generalize to new data (red validation line increases). (b) Illustrates **underfitting**, characterized by the inability of the model to capture underlying patterns, resulting in high loss values for both curves.

Using Keras Tuner, the model was evolved through seven major versions (Table 6). The final rigorous regularization strategy (Version 11) included: an ultra-low learning rate (5×10^{-6}) for stable convergence, a high dropout rate (0.7), strong L2 regularization (0.5 with weight decay 0.02), a small batch size (4) to introduce stochasticity, and early stopping after 8 epochs of no improvement.

Table 6. Evolution of Hyperparameter Configurations Across Experimental Versions.

Version	Learning Rate	Dropout	L2 Reg.	Batch Size	Val Loss Gap	Accuracy (%)
V1 (Baseline)	3×10^{-5}	0.4	0.001	8	N/A	94.7
V2	2×10^{-6}	0.4	0.01	4	0.018	94.3
V3	2×10^{-6}	0.4	0.01	4	0.051	94.8
V4	1×10^{-5}	0.3	0.01	8	0.037	95.8
V5	1×10^{-5}	0.4	0.1	8	0.037	95.8
V6	1×10^{-5}	0.5	0.5	8	0.051	94.8
V11 (Final)	5×10^{-6}	0.7	0.5	4	0.058	95.36

3.3. Stage 3: Production Deployment

The final stage of the framework addresses the deployment gap by implementing the optimized model into a production-ready, containerized web application (Figure 5). This stage is designed to be model-agnostic: the same containerized architecture can host any transformer model produced by Stage 2, regardless of the specific detection domain. A critical design principle was to minimize the technical barrier for end users: the application requires no machine learning expertise, no local model installation, and no command-line interaction—users simply submit a URL through a web browser. This accessibility is essential for the framework’s stated goal of empowering communities with limited technical resources to combat deceptive content on social media.

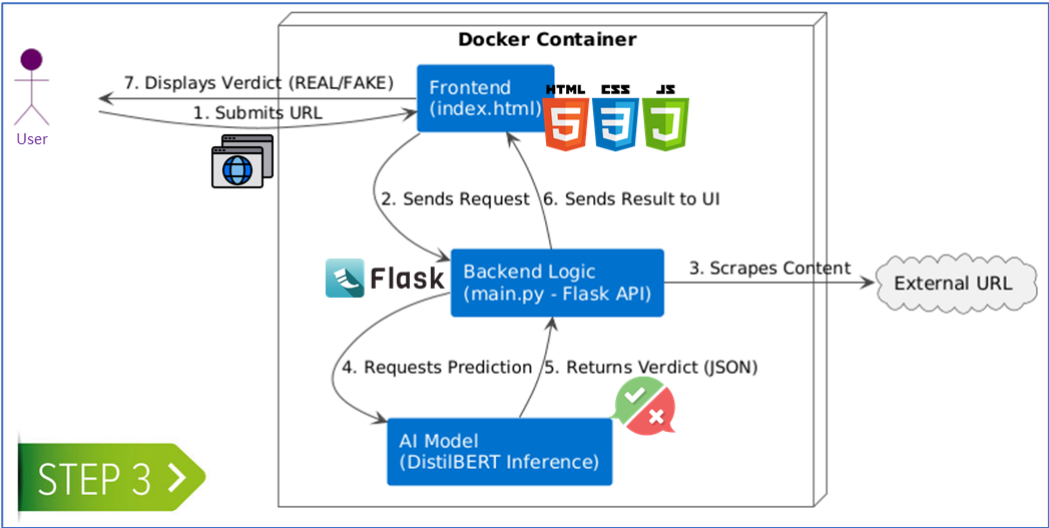


Figure 5. Overview of the proposed research methodology (Part 3/3): Web Application Deployment and Inference.

The interaction flow, from user request to final verdict, follows a well-defined pipeline that ensures both reliability and transparency. When a user submits a URL, the system performs content extraction, text preprocessing, tokenization, model inference, and result presentation in a single automated pass. The complete workflow is detailed in the sequence diagram (Figure 6). Notably, the system provides not only a binary classification (real or fake) but also a confidence score derived from the softmax output layer, enabling users to assess the certainty of each prediction. This probabilistic output is particularly valuable in edge cases where the content may contain a mix of factual and misleading information.

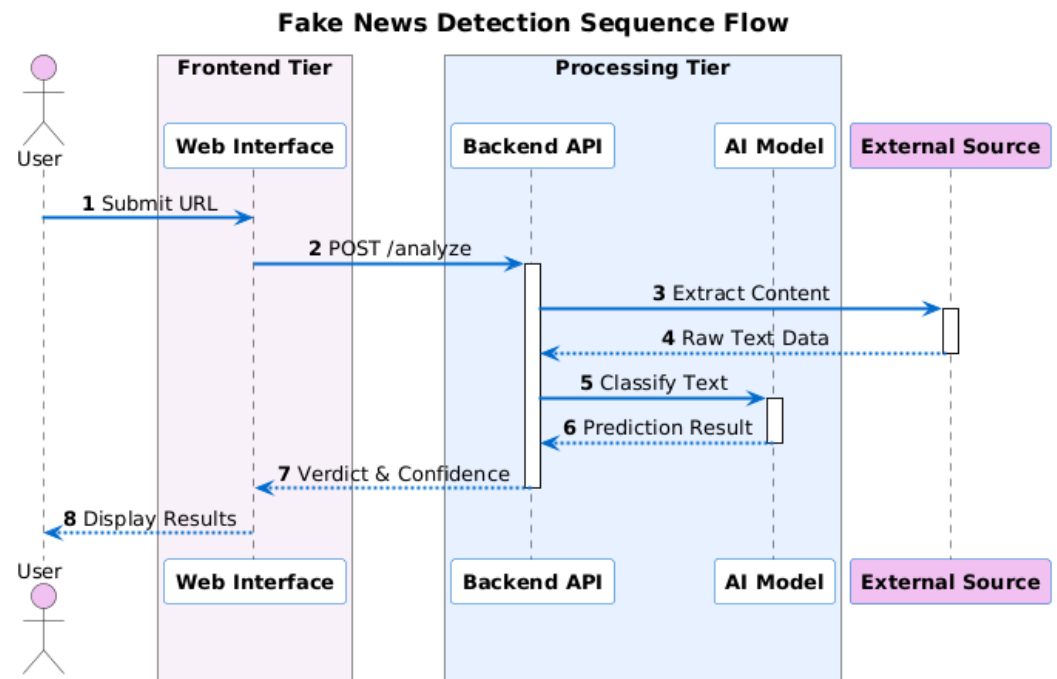


Figure 6. Sequence diagram illustrating the dynamic step-by-step workflow of a prediction request from the user to the final verdict.

The system consists of four integrated components, each designed for modularity and replaceability:

1. **User Interface:** A responsive static web frontend built with HTML, CSS, and JavaScript where users submit URLs for analysis. The interface provides real-time visual feedback during the analysis process—including loading indicators and color-coded results (green for real, red for fake)—to ensure an intuitive user experience for non-technical users.
2. **API Backend:** A Flask-based RESTful service exposing an /analyze endpoint that receives POST requests containing the target URL. The backend implements error handling for common failure scenarios such as unreachable URLs, non-textual content, and pages with insufficient text for meaningful classification.
3. **Inference Engine:** The core analytical component loads the trained DistilBERT model and tokenizer at startup to minimize per-request latency. Upon receiving a URL, it scrapes the page content by extracting `<h1>` and `<p>` tags via BeautifulSoup, applies text cleaning (removing HTML artifacts, normalizing whitespace), tokenizes the combined title and body with a [SEP] separator token, and computes probability scores through a softmax function. The engine truncates inputs exceeding 512 tokens—DistilBERT’s maximum sequence length—ensuring graceful handling of long-form articles.
4. **Docker Deployment:** The entire runtime environment—including Python 3.10 dependencies, PyTorch, the Hugging Face Transformers library, and the serialized model artifacts—is encapsulated in a Docker image built from a minimal base to ensure reproducibility, portability, and ease of deployment across platforms.

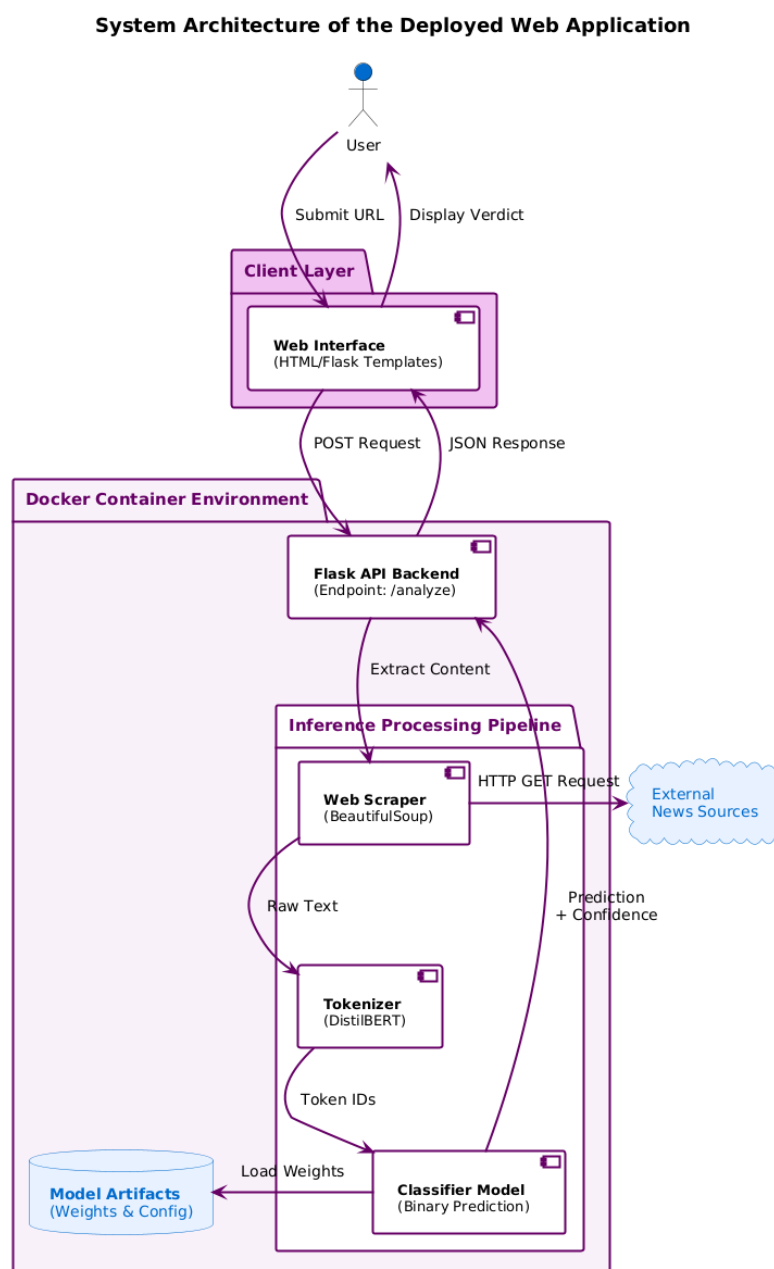


Figure 7. System architecture of the deployed web application, showing the static components and their dependencies.

To guarantee scalability and reproducibility, the system was architected as a containerized microservice stack. This modular approach (Figure 7) separates the frontend interface from the backend inference tasks, allowing for independent scaling and easier maintenance. The containerized design also facilitates continuous integration workflows: when the model is retrained on an updated corpus (e.g., incorporating new forms of deceptive content such as deepfake-related misinformation or emerging scam patterns), only the model artifacts within the Docker image need to be replaced—no changes to the application code, API endpoints, or frontend interface are required. This decoupling between the detection model and the serving infrastructure is a deliberate architectural choice that reinforces the framework's transferability across detection domains. Furthermore, the Docker-based deployment enables rapid replication in diverse hosting environments, from local machines for individual researchers to cloud-based virtual machines for institutional deployments,

ensuring that the framework remains accessible regardless of the user's computational infrastructure.

3.4. Evaluation Metrics

To assess model performance, standard metrics derived from the confusion matrix were employed, which categorizes predictions into true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). In this context, the "positive" class represents real news (label 1), and the "negative" class represents fake news (label 0). These metrics are widely used in fake news detection literature, facilitating comparison with related work.

- **Accuracy:** Represents the overall proportion of correct predictions. While a useful general indicator, it can be misleading in imbalanced datasets.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

- **Precision:** Measures the accuracy of positive predictions. High precision is necessary to minimize the misclassification of fake content as real.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

- **Recall (Sensitivity):** Indicates the proportion of actual real news correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

- **F1-Score:** The harmonic mean of precision and recall. It provides a balanced metric, particularly valuable for uneven class distributions.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

- **Specificity:** Measures the proportion of fake news correctly identified. This metric is crucial for detection systems to ensure deceptive content is accurately flagged.

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (5)$$

For the multiclass comparison, macro-averaged scores were utilized, calculating metrics for each class independently and then averaging them to ensure equitable representation.

4. Results: Framework Validation on Spanish Fake News Detection

4.1. Performance of the Metaheuristic Baseline (Stage 2a)

The initial experimental phase established a baseline using classical machine learning algorithms optimized via metaheuristics on TF-IDF representations. Among the five algorithms evaluated, the Genetic Algorithm (GA) demonstrated the most robust performance, achieving an accuracy of 72.03% and a macro F1-score of 0.714 (Table 8). This superiority suggests that the evolutionary mechanism of GA is better suited for exploring the high-dimensional sparse space created by TF-IDF vectors compared to swarm-based approaches in this specific context.

Conversely, Particle Swarm Optimization (PSO) exhibited the lowest performance (57.67% accuracy) and failed to converge effectively. This divergence highlights a critical limitation of classical approaches: their heavy reliance on keyword frequency (TF-IDF)

lacks the semantic understanding necessary to detect sophisticated fake news, resulting in high variance between optimization strategies. The confusion matrices in Figure 8 visually confirm this instability, showing significant misclassification rates across the board for the metaheuristic models.

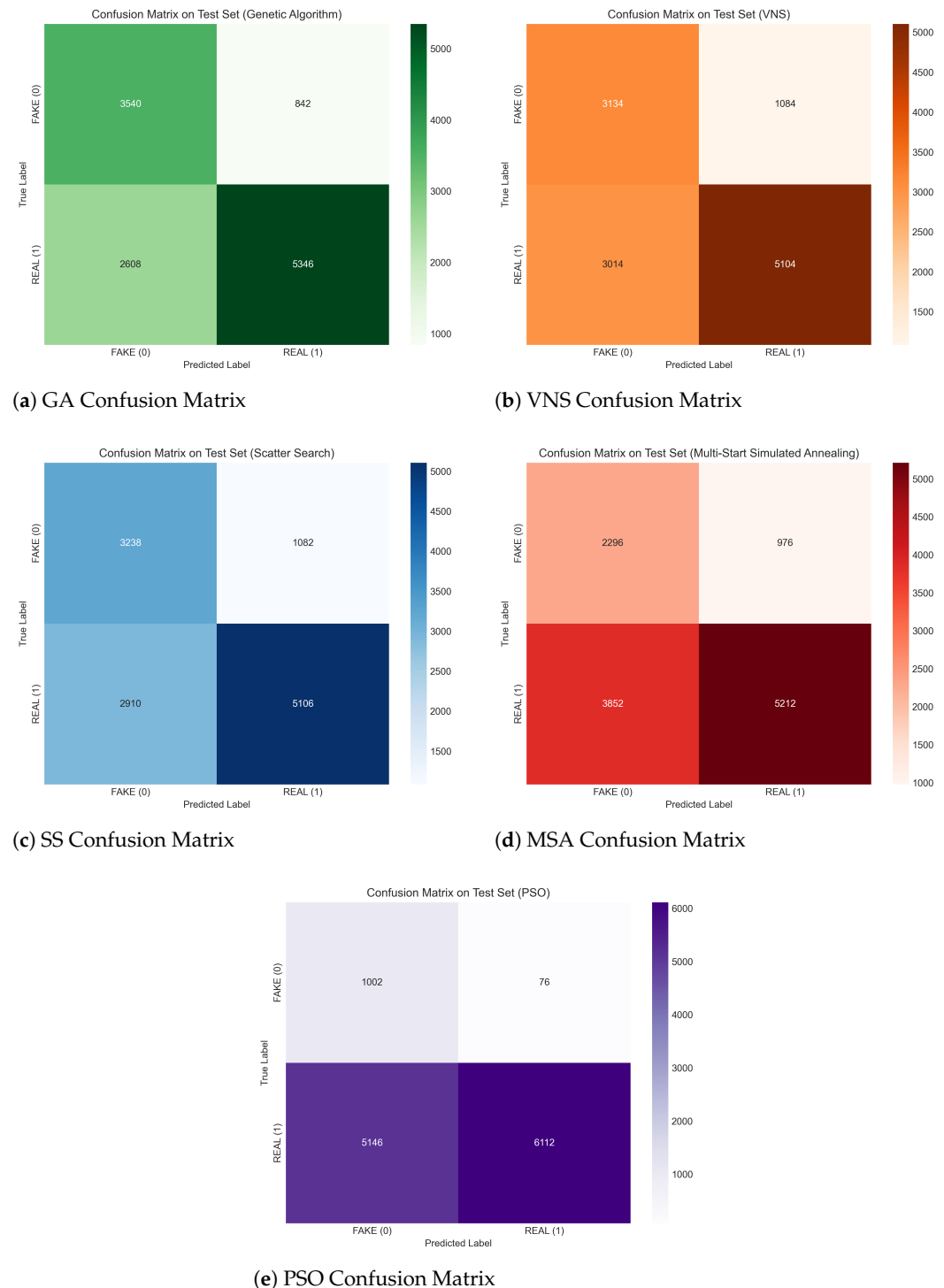


Figure 8. Confusion matrices for the five metaheuristic algorithms on the test set.

4.2. Performance of the Systematically Fine-Tuned Transformer (Stage 2b)

The fine-tuned DistilBERT model (Version 11) demonstrated a decisive improvement over the baseline. As detailed in Table 7, the model achieved an accuracy of 95.36% on the

test set. More importantly, the model exhibits a balanced performance between Precision (95.4%) and Recall (95.4%).

In the context of fake news detection, **Specificity** (94.5%) is a critical metric, as it represents the model’s ability to correctly identify actual fake news (True Negatives in the adopted configuration) without flagging legitimate news as false. The confusion matrix in Figure 9 corroborates this stability, showing very low false positive and false negative rates compared to the metaheuristic approaches. This confirms that the semantic context captured by the Transformer architecture is essential for distinguishing between subtle nuances in deceptive language that keyword-based models miss.

Table 7. Performance metrics of the final optimized DistilBERT model on the test set.

Metric	Value (%)
Accuracy	95.36
Precision	95.4
Recall (Sensitivity)	95.4
F1-Score	95.35
Specificity	94.5

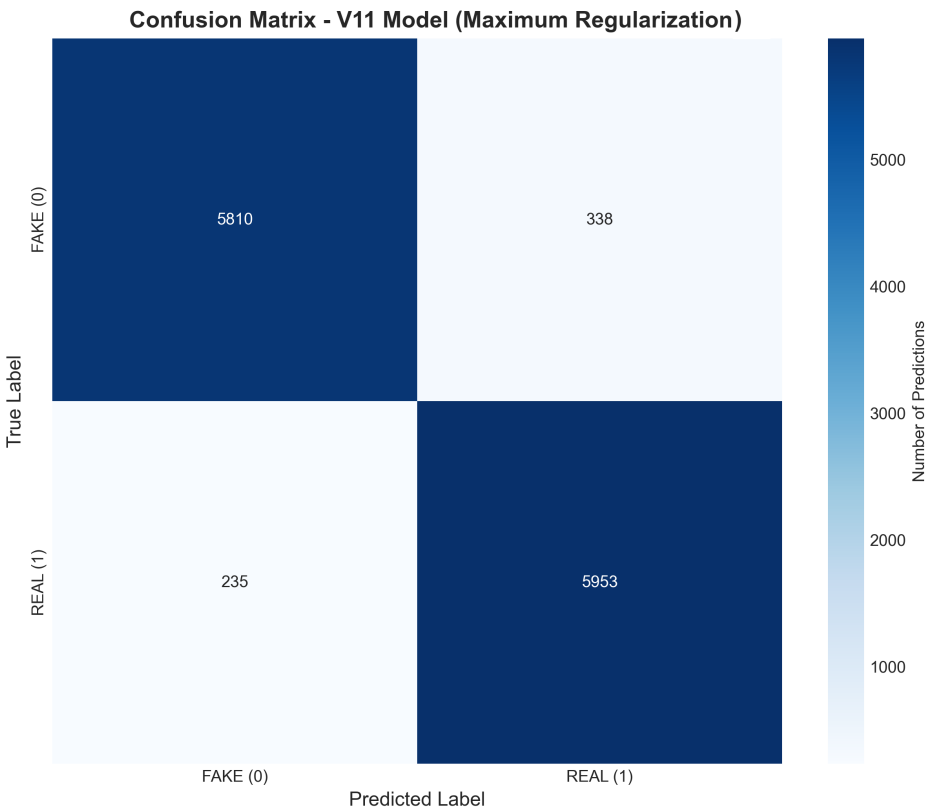


Figure 9. Confusion matrix for the final DistilBERT model on the test set.

4.3. Final Comparative Analysis: Metaheuristics vs. Transformer

To quantify the impact of the architectural shift from classical ML to Deep Learning, a direct comparison of the best-performing models from each paradigm was performed. Table 8 presents the consolidated results.

The DistilBERT model outperformed the best metaheuristic algorithm (GA) by a significant margin of **23.33 percentage points** in accuracy (95.36% vs. 72.03%). This disparity illustrates the "semantic gap": while metaheuristics can optimize the decision boundary for keyword frequencies, they cannot compensate for the lack of contextual

understanding inherent in TF-IDF representations. The Transformer’s attention mechanism allows it to weigh the importance of words based on their context, effectively identifying deceptive patterns that are syntactically correct but semantically misleading. Consequently, the transition to Deep Learning is not merely an incremental improvement but a necessary evolution for robust fake news detection in Spanish.

Table 8. Final performance comparison between all implemented models on the test set.

Algorithm	Accuracy (%)	F1-Score (macro)	Precision (macro)	Recall (macro)	Specificity (%)	Ranking
Transformer-Based Approach						
DistilBERT (Final)	95.36	0.954	0.954	0.954	94.5	1st
Metaheuristic-Optimized Classical Approaches						
Genetic Algorithm (GA)	72.03	0.714	0.740	0.720	57.6	2nd
Scatter Search (SS)	67.64	0.669	0.693	0.676	52.7	3rd
VNS	66.78	0.659	0.686	0.667	51.0	4th
Simulated Annealing (MSA)	60.86	0.586	0.638	0.608	37.4	5th
Particle Swarm Opt. (PSO)	57.67	0.489	0.736	0.575	16.3	6th

4.4. Overfitting Control Analysis

The implementation of a rigorous regularization strategy successfully mitigated overfitting. Training concluded after 23 epochs due to early stopping, with the optimal checkpoint identified at epoch 17. At this point, training accuracy was 98.6% and validation accuracy was 95.36%. Figure 10 illustrates the evolution of accuracy and loss for the final model (V11).

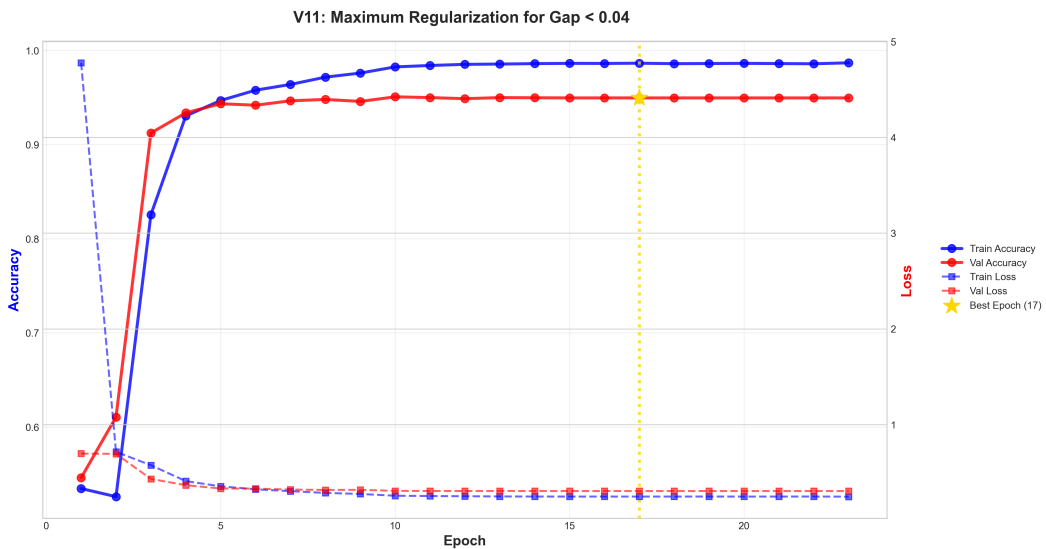


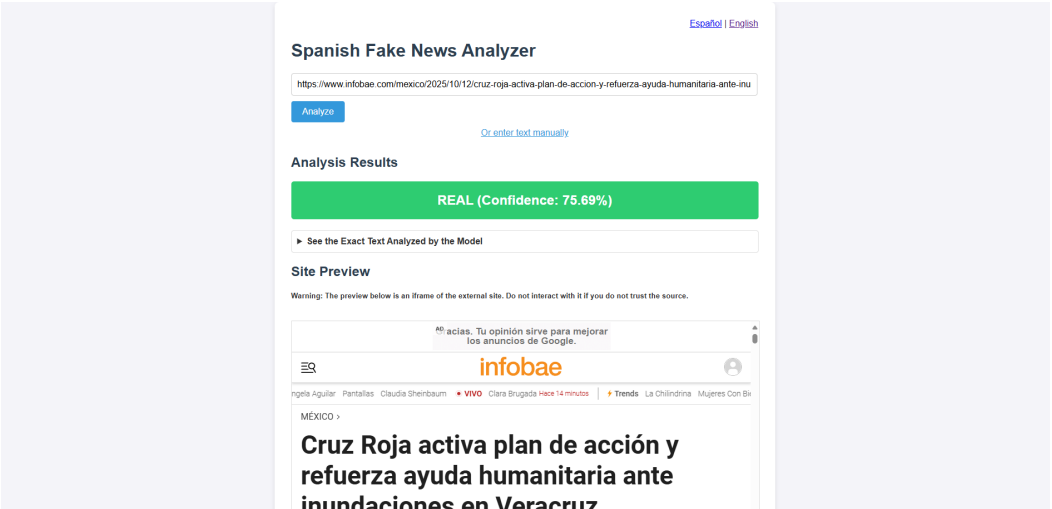
Figure 10. Evolution of performance metrics during the fine-tuning of the final model (V11). The x-axis represents the training epochs, while the y-axes represent Loss (left) and Accuracy (right). The blue lines denote training performance, and the red lines denote validation performance. The gold star identifies the optimal checkpoint at Epoch 13, where the generalization gap (0.058) was minimized before the onset of overfitting.

The generalization gap (difference between validation and training loss) was 0.058, approaching the target of <0.04 and remaining well below the 0.10 threshold typically indicating overfitting.

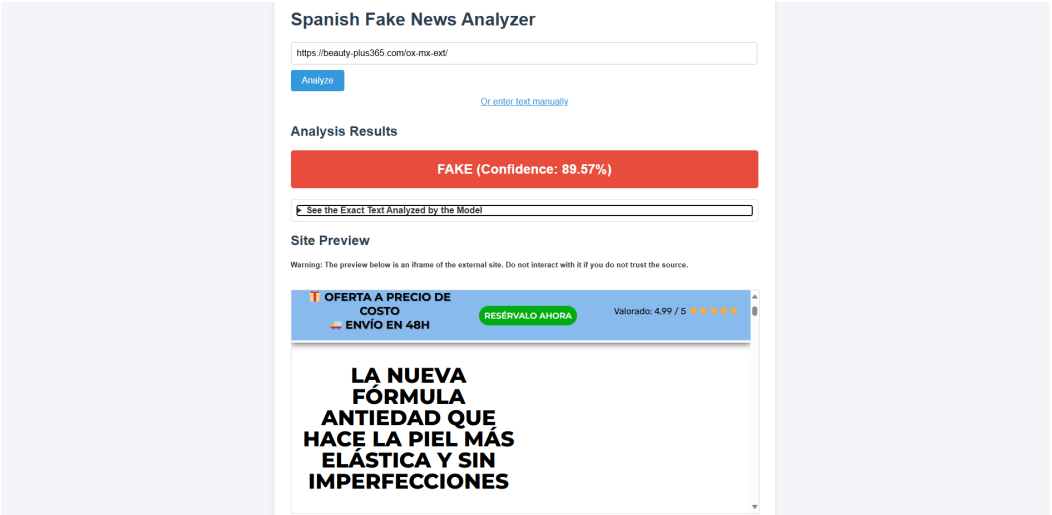
4.5. Deployment Validation (Stage 3)

The deployed web application demonstrated robustness in real-world scenarios. It correctly identified various types of content, including authentic news, fabricated stories, and investment scams, providing preliminary evidence that the framework is

transferable to other forms of deceptive content on social media beyond traditional fake news. Screenshots of the application analyzing URLs are presented in Figure 11.



(a) Correctly identifying a real news article with 93.59% confidence.



(b) Successfully identifying a misleading fake news piece with 94.31% confidence.

Figure 11. Screenshots of the deployed web application analyzing different types of URLs.

5. Discussion

This study proposed a systematic and transferable framework for fine-tuning pre-trained transformer models for domain-specific deceptive content detection on social media, validated through a comprehensive case study on Spanish-language fake news. The transition from classical metaheuristic optimization to the proposed regularization strategy for DistilBERT yielded a 23.33 percentage point improvement in accuracy. Beyond the raw metrics, it is crucial to analyze the framework’s strengths, the model’s boundaries, and the potential for domain transfer.

5.1. Strengths of the Proposed Framework

The primary technical contribution lies not in the final accuracy figure, but in the systematic and reproducible nature of the optimization process. By documenting the full hyperparameter trajectory (Table 6)—including failed configurations—this work provides a practical blueprint that practitioners can adapt for their own detection tasks. The aggressive

regularization strategy (high dropout of 0.7 and strong L2 regularization of 0.5) was identified as critical for preventing overfitting when fine-tuning on domain-specific corpora of moderate size, a challenge that is universal across detection domains, not specific to fake news.

The resulting DistilBERT model demonstrates strong semantic robustness. Unlike the TF-IDF baseline, which relies on keyword frequency and struggles with sarcasm or subtle linguistic cues, the transformer model effectively captures contextual dependencies. This is evidenced by its high Specificity (94.5%), indicating that the model is exceptionally strong at distinguishing deceptive content without aggressively flagging legitimate content—a common pitfall in automated moderation systems. The generalization gap of only 0.058 confirms that the regularization strategy successfully addressed overfitting.

5.2. Transferability to Other Social Media Deception Domains

A central claim of this work is that the proposed three-stage framework is transferable beyond fake news detection. The architecture of the pipeline—corpus unification, systematic regularization, containerized deployment—is explicitly domain-agnostic. To illustrate this transferability, consider the following concrete scenarios where the same framework could be applied by substituting only the training corpus:

- **Investment Scam Detection:** Social media platforms, particularly Facebook and Instagram, are inundated with fraudulent pages promoting fake investment opportunities (e.g., petroleum investments, cryptocurrency schemes). These pages use persuasive language patterns that share rhetorical features with fake news—urgency, emotional manipulation, and fabricated testimonials. A concrete example is a page impersonating PEMEX (Mexico’s state oil company) that claims citizens can “earn 100,000 pesos” by joining a fictitious “social project” [6]—a classic social engineering tactic combining authority impersonation with unrealistic financial promises. A corpus of labeled scam pages could be constructed using the Stage 1 unification methodology, and the Stage 2 regularization recipe applied directly to fine-tune a detection model.
- **Fraudulent E-Commerce Detection:** Fake retail pages impersonating legitimate brands or promoting non-existent products represent a growing threat that disproportionately affects elderly and digitally illiterate populations. Real-world examples include pages selling fraudulent “anti-aging cosmetics” using fabricated before-and-after photos, fake WhatsApp testimonials, and artificial urgency tactics (e.g., countdown timers and “order now” buttons) [7], as well as pages advertising implausible products such as “smart radiation-proof reading glasses” claiming to sell over 100,000 units per day [8]. The textual patterns of these pages—exaggerated claims, pressure tactics, and suspicious product descriptions—are amenable to the same transformer-based classification approach.
- **Phishing Content Detection:** Phishing campaigns disseminated through social media messaging share linguistic features with misinformation: impersonation of authority, urgency, and deceptive intent. The framework’s containerized deployment stage (Stage 3) is particularly valuable here, as it enables real-time URL analysis that could be extended to phishing detection with minimal architectural changes.

The use of distilbert-base-multilingual-cased as the base architecture further enhances transferability, as the same pretrained model supports over 100 languages. Extending the framework to a new language primarily requires the curation of a language-specific corpus (Stage 1) rather than a redesign of the regularization strategy (Stage 2) or the deployment architecture (Stage 3).

5.3. Limitations

Despite its high performance, the model is not infallible. A key limitation is its static nature; it was trained on a closed corpus ending in early 2025. Consequently, the model is susceptible to *concept drift*, where it may fail to detect novel misinformation narratives or evolving linguistic tactics used in future disinformation campaigns. Additionally, the current approach is strictly unimodal (text-only). It fails when the deception is embedded in multimedia elements or when analyzing extremely short texts like tweets or headlines where semantic context is scarce. Furthermore, while the framework is designed to be transferable, the empirical validation presented here is limited to a single domain (Spanish fake news); future work should validate the framework on additional deception domains to strengthen the transferability claim.

5.4. Future Work

Future work will focus on three critical areas to address these limitations and expand the framework's utility. First, the implementation of Continuous Learning (CL) pipelines is envisioned to update the model with fresh data periodically without catastrophic forgetting, mitigating the issue of concept drift. Second, the framework will be empirically validated on additional social media deception domains—specifically, a corpus of investment scams and fraudulent e-commerce pages will be constructed to demonstrate cross-domain transferability. This will involve applying the same three-stage pipeline to these new domains and comparing the resulting models' performance. Third, the expansion of the web application's API to support browser extensions is planned, bringing the detection capability directly to the user's browsing experience and enabling real-time protection against multiple forms of social media deception simultaneously.

6. Conclusions

This study presents a systematic and transferable framework for fine-tuning pretrained transformer models for domain-specific deceptive content detection on social media. The framework is structured as a three-stage pipeline—corpus unification, systematic regularization, and containerized deployment—designed to be reusable across languages, content domains, and social media platforms.

The framework was validated through a comprehensive case study on Spanish-language fake news detection. A unified corpus of 61,674 articles was constructed from four heterogeneous sources, demonstrating the corpus unification methodology. Through a rigorous experimental process involving over 500 GPU hours, the proposed aggressive regularization strategy for DistilBERT achieved 95.36% accuracy with a generalization gap of only 0.058, significantly outperforming metaheuristic-optimized classical baselines by 23.33 percentage points. The full optimization trajectory—including failed configurations—is documented to provide a reproducible recipe for practitioners.

Beyond the case study, the framework's core technical contribution lies in its transferability. The same three-stage pipeline can be applied to detect other forms of social media deception—including investment scams, fraudulent e-commerce pages, and phishing content—by substituting the training corpus. The deployment of the optimized model into a Dockerized web application demonstrates that the framework extends beyond academic benchmarks to deliver practical tools for digital self-defense. The open-source availability of the entire framework—corpus, model, and application—facilitates further adaptation, paving the way for more resilient detection systems that can protect vulnerable populations against the evolving landscape of social media deception.

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